

Zihao Zhang

School of Cyber Science and Technology, USTC

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EDUCATION

University of Science and Technology of China

Bachelor of Cyber Science and Technology

Anhui, China

2022 - Present

- Overall GPA: 3.80/4.3 Rank: **4th**/71 (5.6%)

RESEARCH INTERESTS

Cryptography, Applied Cryptography, Zero-Knowledge Proofs

PUBLICATIONS

[1] Certus, Efficient Verifiable Convolutional Neural Network Inference based on Zero-Knowledge Proofs

(Submitted to Eurocrypt 2026)

Wenjie Qu, **Zihao Zhang**, Yanpei Guo, Jiaheng Zhang

[2] Efficient SNARK for Floating-Point Computation

(Submitted to IEEE S&P 2026)

Wenjie Qu*, **Zihao Zhang*(co-first author)**, Jiaheng Zhang

RESEARCH EXPERIENCE

Efficient Zero Knowledge Proofs for Convolutional Neural Networks (CNNs)

Mar 2025 – July 2025

Research Assistant at NUS. Advisor: Prof. Jiaheng Zhang <[Google Scholar](#)>, Dr. Wenjie Qu

- Participated in deriving efficient prover-friendly constraints for checking multi-channel 2D convolutions in CNN by making use of relationship between matrix convolution and mathematical 2D convolution.
- Completed soundness analysis of the protocol.
- Implemented the framework using C++.
- Tested the protocol on VGG-16 and ResNet-50 and achieved obvious speedup over prior works, such as Modular (CCS'23), zkCNN (CCS'21) and zkNN (eprint'24).
- Contributed to writing and editing of the manuscript.

Efficient Zero Knowledge Proofs for Floating-Point Computations

Aug 2025 – Nov 2025

Research Assistant at NUS. Advisor: Prof. Jiaheng Zhang <[Google Scholar](#)>, Dr. Wenjie Qu

- Designed a mapping for unifying different numbers in the IEEE-754 standard in the circuit to make them more circuit-friendly.
- Completed analysis of edge cases for addition, multiplication and division.
- Derived decomposing scheme of exp, ln, sin and cos by leveraging their mathematical structures, which was combined with per-chunk lookup scheme to obtain an efficient ZKP framework for proving these transcendental functions.
- Implemented Spartan, Liger0, LogUp and our SNARK in C++, and tested our SNARK.
- Contributed to writing and editing of the manuscript.

ACADEMIC PROJECTS

Tiny-C Compiler (Course Project for Compiler Theory B)

Sep 2024 – Dec 2024

Advisor: Prof. Weihai Li

- Developed a simplified C compiler in C, implementing lexical analysis, syntax analysis, and Intermediate Representation (IR) generation for the Tiny-C language.

TEACHING ASSISTANT EXPERIENCE

Computer Programming

Sep 2024 – Jan 2025

Advisor: Prof. Jian Tang

SELECTED AWARDS AND HONORS

- Outstanding Student Scholarship, Golden Award (**top 5%**) Oct 2023
- Wang Xiaomo **Talent Program** in Cyber Science and Technology Aug 2023 – July 2024
- Scholarship of Wang Xiaomo Talent Program Jan 2024
- Huawei Scholarship (**1 winner among 73 students**) Oct 2024

RELEVANT COURSES

Computer Science: C Programming, Data Structures and Algorithms, Operating Systems, Computer Networks, Algorithms Analysis and Design, Cryptography, Compiler Theory

Mathematics: Mathematical Analysis, Linear Algebra, Foundation of Algebra, Probability Theory and Mathematical Statistics, Information Theory

SKILLS

Programming Languages: C/C++, Rust, Python, HTML, CSS, JavaScript

Miscellaneous: SQL, Linux, Git, LaTeX, Markdown, Torch

English: TOEFL: 106 (Reading: 29; Listening: 24; Speaking: 25; Writing: 28)

EXTRACURRICULAR ACTIVITIES & INTERESTS

Member of Student Union of School of Cyber Science and Technology

Sep 2023 – Jun 2024

- Organized a campus-wide badminton tournament as the person in charge. I was responsible for all tasks except financial reimbursement and event promotion

Independent Server Administration & Configuration Experience

Dec 2024 – Aug 2025

- Deployed and configured a stable Ubuntu 20.04 LTS environment on a Raspberry Pi 4B independently from scratch, and built a Forge-based Minecraft server and installed mods and plugins such as *Create* and *LuckPerms*.
- Implemented a reliable NAT traversal setup using FRP to enable secure remote access.